WaveOrder for label-free high-throughput screening

This guide demonstrates WaveOrder for label-free high-throughput screening using optical pooled screening (OPS) data from the Biohub's multimodal perturbation atlas [1], where phase from defocus enabled label-free tracking and phenotyping of CRISPR-perturbed A549 cells. Brightfield defocus stacks from a standard motorized widefield microscope, together with a WaveOrder config [2], yield quantitative phase at screening scale (Figure 1).

After reading this guide and acquiring data, use the first-reconstruction guide and connect your agent to the Biohub MCP to run the `waveorder-tutorial` skill. Methods are described in the WaveOrder preprint [3].

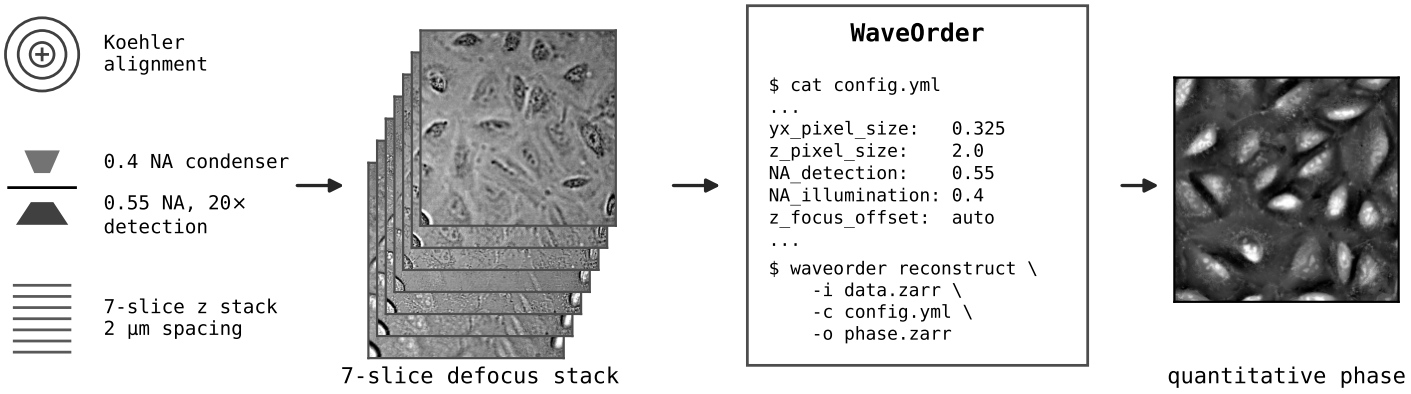


Figure 1. Key acquisition settings (left) produce a defocus stack with defocus-dependent contrast that WaveOrder converts into quantitative phase.

Acquisition

The three settings that matter most are:

- **Koehler illumination.** Align the condenser before every session. Misalignment is the most common source of artifacts. Ron Vale's five-minute iBiology tutorial [4] is a good refresher.
- **Condenser NA ≈ 2/3 of detection NA.** We used a 0.4 NA condenser and a 0.55 NA objective. A larger condenser aperture reduces contrast, while a smaller condenser aperture reduces contrast and resolution.
- **Z spacing and range.** At 20×, start at 2 µm spacing and cover the full cell height with visibly blurred features on either end of the stack (here, 7 slices = 12 µm). For different objectives, scale inversely with magnification: e.g. 4 µm at 10×, 1 µm at 40×.

Parameter	Perturbation atlas value	Notes
Objective	20×, 0.55 NA, air	any objective works; match the config
Condenser	0.4 NA	≈2/3 of detection NA, Koehler aligned
Z stack	7 slices, 2 µm	confirm defocused features on top and bottom slices; use more slices for thicker samples
Camera	6.5 µm pixels	6.5 µm / 20× = 0.325 µm at the sample
Wavelength	450 nm	any wavelength is fine; ideally use a 10 nm notch filter
Exposure	a few ms	fill 1/2–2/3 of the dynamic range

Table 1. Acquisition parameters for the Biohub's multimodal perturbation atlas dataset.

Reconstruction

Convert to OME-Zarr with iohub (`iohub convert -i <dataset> -o data.zarr`), record the acquisition parameters in a config, reconstruct one position, and compare against the raw data in napari:

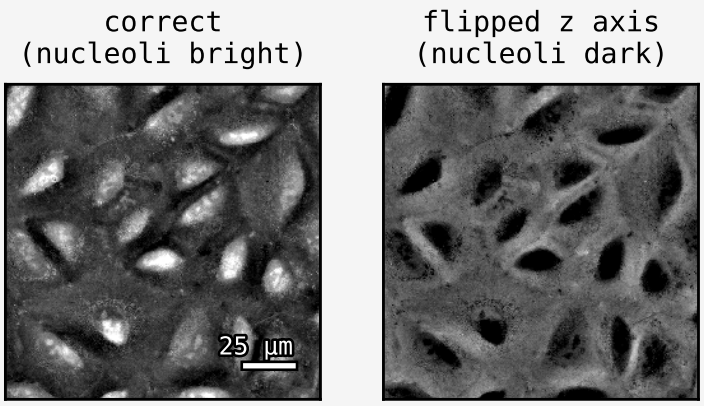
```
waveorder reconstruct \
    -i data.zarr/A/1/029029 \
    -c config.yml -o phase.zarr
waveorder view data.zarr/A/1/029029 \
    phase.zarr/A/1/029029
```

Dense structures (nucleoli, nuclear envelopes, cell margins) should appear brighter than background, as in Figure 1. `reconstruction_dimension: 2` returns one phase image per defocus stack, the best choice for samples that are thin compared to the depth of field of the microscope. If the sample appears to blur all at once as you defocus, use 2. If different regions of the sample blur at different defocus positions, consider trialling 3.

```
$ cat config.yml

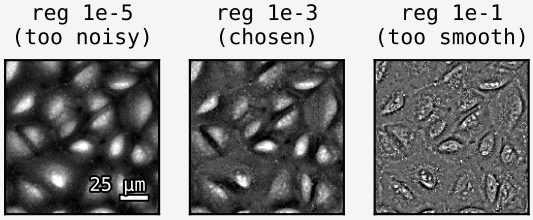
input_channel_names: [BF]
reconstruction_dimension: 2
phase:
  transfer_function:
    yx_pixel_size: 0.325  # 6.5 um / 20x
    z_pixel_size: 2.0
    wavelength_illumination: 0.450
    index_of_refraction_media: 1.0
    numerical_aperture_detection: 0.55
    numerical_aperture_illumination: 0.4
    invert_phase_contrast: false
    z_focus_offset: auto
  apply_inverse:
    regularization_strength: 0.001
```

Failure mode 1 | Flipped z axis



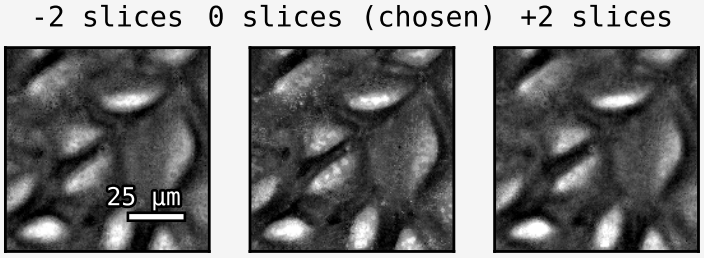
Stages disagree on the z sign convention—is the first slice above or below focus? If the convention is flipped, contrast will be inverted and downstream comparison will be challenging. Dense organelles should be bright. Fix: `invert_phase_contrast: true`. Check once per microscope by reconstructing one stack both ways.

Failure mode 2 | Incorrect regularization



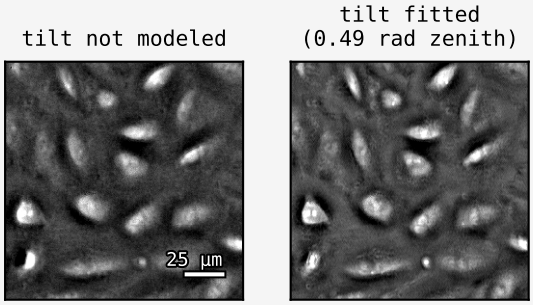
`regularization_strength` balances noise and resolution and varies by orders of magnitude across samples and exposures. Sweep decades on one position and choose manually. Too small amplifies blotchy low-frequency noise; too large leaves a washed-out image. The optimum is usually within a decade of `1e-3`.

Failure mode 3 | Autofocus drift



When the stack center (e.g. the 4th slice in a 7 slice stack) is not in focus, the reconstruction will be blurry. Setting `z_focus_offset: auto` runs the search that can recover the defocus offset.

Failure mode 4 | Tilted illumination at the well edge



Misaligned Koehler and imaging near a well-edge meniscus tilts the illumination, making cells look shadow-cast with reduced contrast. Fit the tilt by gradient descent in the same config:

```
    z_focus_offset: auto
    tilt_angle_zenith:  {init: 0.13, lr: 0.05}
    tilt_angle_azimuth: {init: 0.0,  lr: 0.05}
optimization: {max_iterations: 50, method: adam}
```

Scaling up

A single config file should generate good reconstructions across a screen. We suggest parallelizing `waveorder reconstruct` over positions (one process or cluster job per position) with `device: cuda:0` to reconstruct on a GPU. We recommend `z_focus_offset: auto` to find the focus of each position.

We recommend manual supervision to set up the optimized `tilt_angle_{zenith,azimuth}` parameters across entire wells. We find that initializing these tilt parameters at good initial tilts generates the stablest reconstructions, so we recommend inspecting reconstructions at the well edges, then refining the initial values until you reach stable reconstructions that pass QC.

References

1. C. Liu, A. V. Hillsley, M. Sekhar, et al., “A Multimodal Perturbation Atlas Defines the Phenotypic Resolution of Cellular Morphology,” bioRxiv 2026.06.01.728087.
2. WaveOrder repository: github.com/biohub/waveorder.
3. T. Chandler, I. E. Ivanov, G. Sturm, et al., “WaveOrder: A differentiable wave-optical framework for scalable biological microscopy with diverse modalities,” arXiv:2412.09775.
4. R. Vale, “Microscopy: Setting Up Koehler Illumination,” iBiology Microscopy Series.